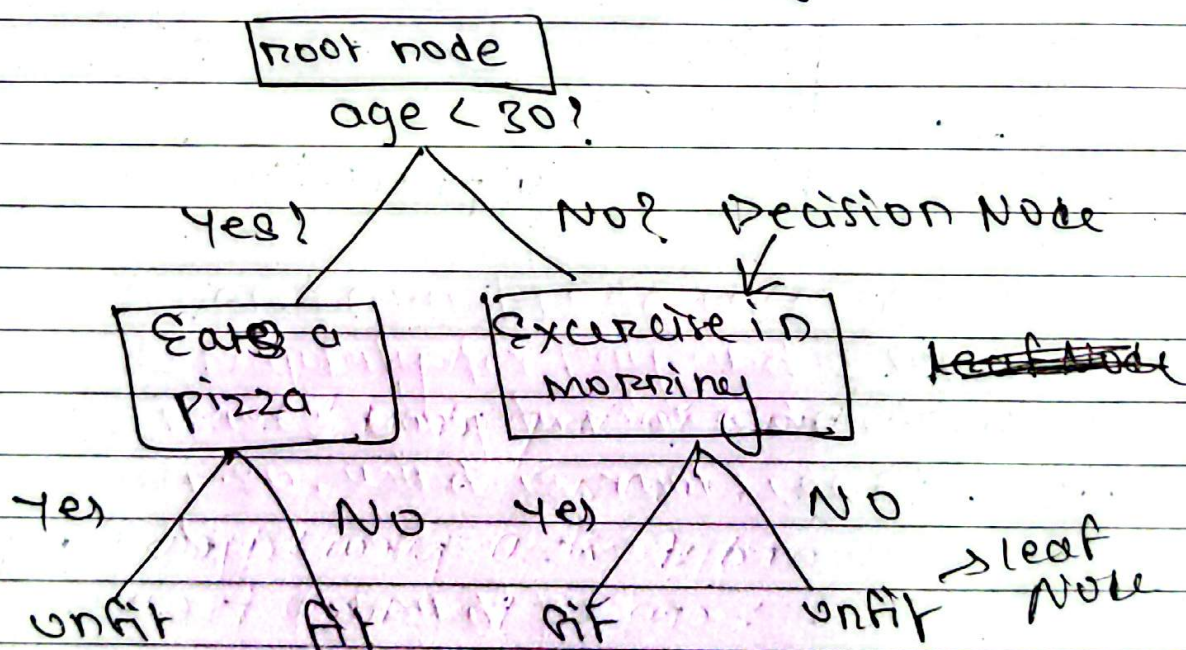


* Decision Tree in ML %

* It is a flowchart which break down dataset into smaller subsets on the basis of specific features. It consist 3 core structure 1) root node 2) Decision node 3) leaf Node

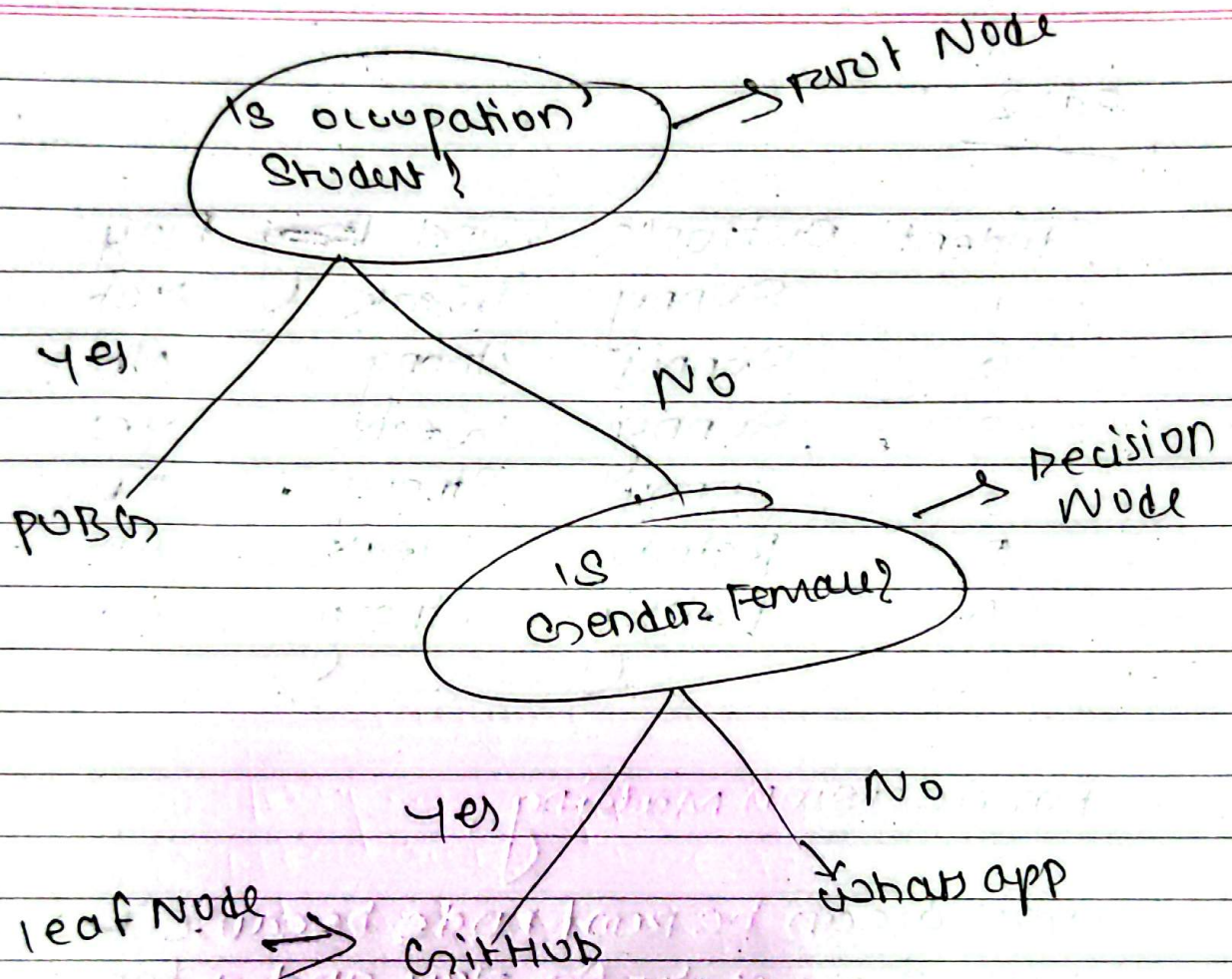
Advantage: It is used for both classification and regression.

% It also uses if else condition for decision structure making.



eg:

Gender	Occupation	Suggestion	
F	Student	pubG	If occupation = 2 Student print (pubG)
F	programmer	GitHob	
M	"	whatsapp	else if occupation = 2 Fem print (GitHob)
F	"	GitHob	
M	Student	pubG	else print (whatsapp)
M	"	"	



Entropy :- It means randomness. ~~measure of~~
 measure of disorder.
 Data में एन्ट्रॉपी का अर्थ

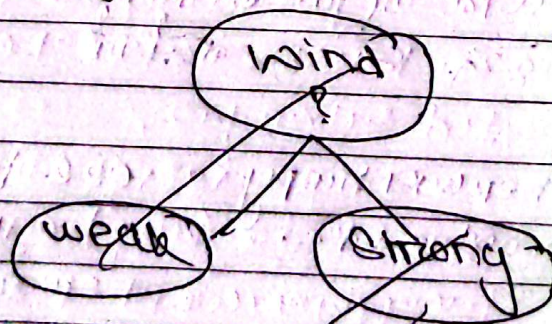
root node → where entropy is less that is kept
 as root node.

eg: 2

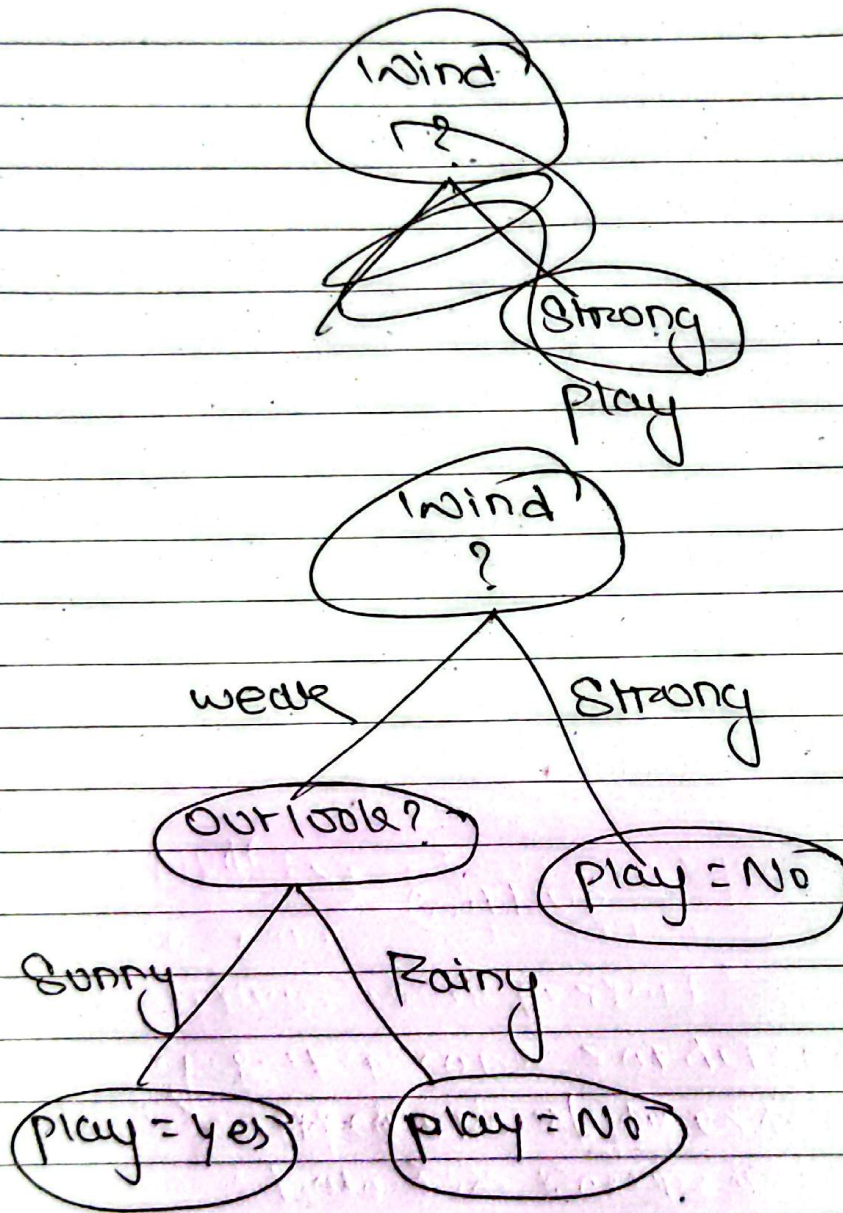
Student	Outlook	Wind	Play
1	Sunny	Weak	Yes
2	Sunny	Strong	No
3	Sunny	Weak	Yes
4	Sunny	Strong	No
5	Rainy	Weak	No
6	Rainy	Strong	No

For decision making,

Wind can be root node because weak has 2 majority for play and 1 for Not play which makes easy to make decision for playing. So, wind is root node.



And if we see Strong it has all No for playing ~~with~~ by which we can directly make decision that when wind is strong we cannot play.



Entropy and information gain are inverse to each other

Note: Entropy is higher then Information gain is less
 " " " " " " " high.